# FINAL YEAR PROJECT PROPOSAL

## AI-Based Cotton Leaf Curl Virus Detection and Treatment Recommendation System Using Deep Learning

### 1. Proposal Title

The AI-Based Cotton Leaf Curl Virus Detection and Treatment Recommendation System Using Deep Learning

### 3. Project Brief

This project aims to build an AI system that can spot Cotton Leaf Curl Virus in cotton plants by looking at pictures of their leaves. The system will use deep learning methods to sort each uploaded or captured leaf picture into one of two groups: Healthy Cotton Leaf or Cotton Leaf Curl Virus. This tool will help farmers spot signs of Cotton Leaf Curl Virus early and give them advice, on how to manage and prevent Cotton Leaf Curl Virus. An Android app will give an interface for users to take a picture or upload a cotton leaf image and then see what the model says about Cotton Leaf Curl Virus.

### 4. Project Description

Cotton is a farm crop in Pakistan. Cotton Leaf Curl Virus is a disease that hurts cotton production. Finding disease signs early can help farmers get advice and act quickly.

The planned system will use a labeled set of pictures that show cotton leaves and cotton leaves with Cotton Leaf Curl Virus. A deep learning tool, best with EfficientNetB0 and transfer learning will be. Tested to sort the pictures.

The taught model will be put into an Android app made with Kotlin and Android Studio. Users can take a photo with their phone camera. Pick a picture from their gallery. The app will look at the picture. Show the predicted class along, with basic disease-management tips.

The model will be judged with accuracy, precision, recall, F1-score and a confusion matrix. The picture set will be reviewed for image quality, correct labels and enough examples of each class before teaching the model.

The system will give image-based guesses, not a final diagnosis. Treatment ideas will be basic. When needed will point farmers to local agricultural experts.

### Main Features

Capture cotton leaf images using a camera.

Upload leaf images from the phone gallery.

Classify each image as Healthy Cotton Leaf or Cotton Leaf Curl Virus.

Show the predicted class. Include general guidance related to the result.

Give information on how to prevent the disease.

Evaluate how well the model performs using classification metrics.

## 5. Technologies

•Programming Language: Python

•Deep Learning: TensorFlow and Keras

•Model: EfficientNetB0, with Transfer Learning

•Image Processing: OpenCV

•Mobile Application: Android Studio and Kotlin

•Dataset: Labelled cotton leaf images containing healthy and Cotton Leaf Curl Virus classes

•Model Evaluation: Scikit-learn and Matplotlib

•Backend: FastAPI

•Version Control: Git and GitHub

## 7. Expected Outcome

I expect to deliver a functional Android application that works with a trained deep learning model. This deep learning model will look at cotton leaf images. Decide if they are healthy or if they show Cotton Leaf Curl Virus. The project will show how artificial intelligence can help farmers spot disease early which is an use of tech in farming. When we finish we will share the results from evaluating the model; we will not guess an accuracy, before testing.